

2-⑥

(II)

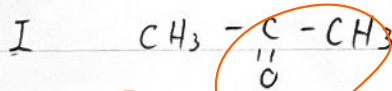
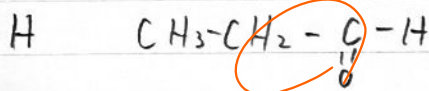
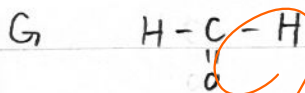
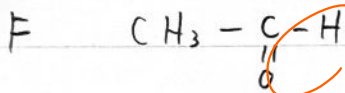
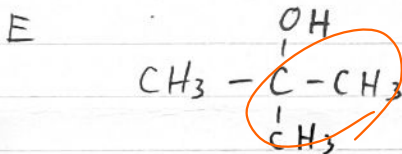
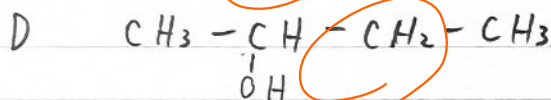
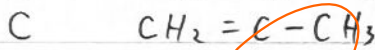
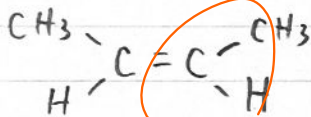
1、ア 酢酸化銅; 試料の不完全燃焼を防ぐ

イ 塩化カルシウム; 水を吸収する

ウ ソーダ石灰; 二酸化炭素を吸収する

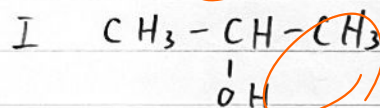
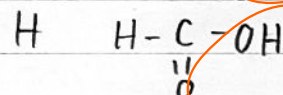
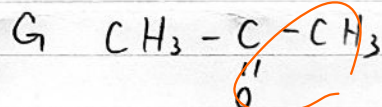
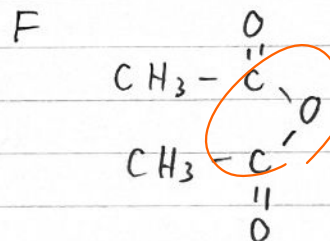
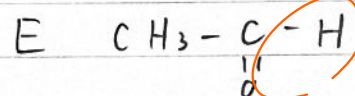
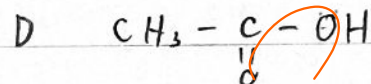
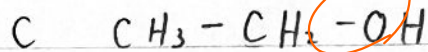
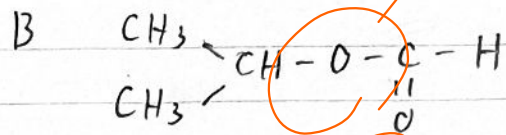
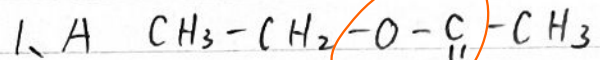
2、 C_4H_8

3、A



4、F

3-⑭



2、ア 水酸化ナトリウム 1ヨウ素

ヨードホルム反応; CHI_3

3、銀鏡反応

4、アセトアルデヒド

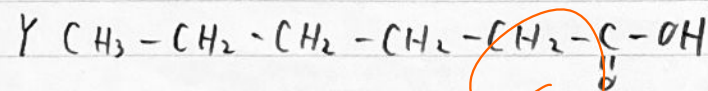
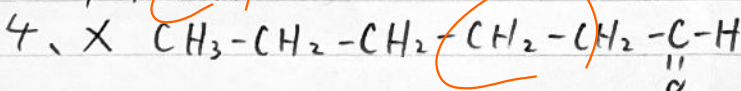
3-③

1、(a) 8種類

(b) 6種類

2、3種類

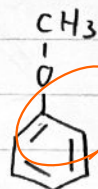
3、4種類



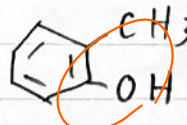
4-③

1、 C_7H_8O 2、A CH_2OH 

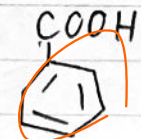
B



C

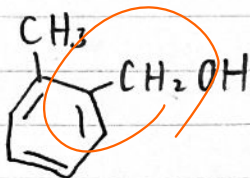


D

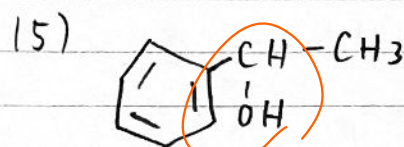
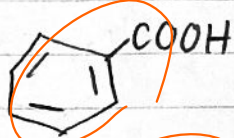


4-14

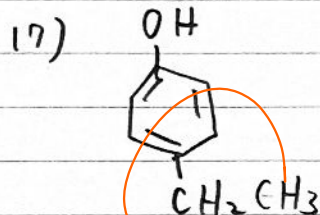
(1) 4 (2)

(3) CHI_3

(4)



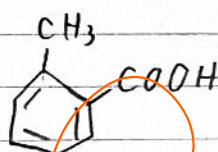
(6) 9



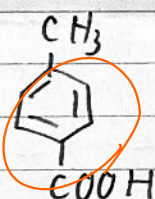
(18) 5 (19) 1:2:2:2:3

4-16

1. (1) A

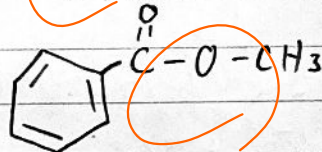


B

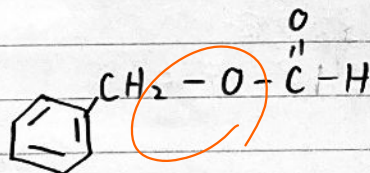


(2) o-キシレン、ナフタレン

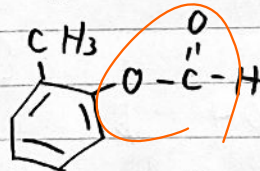
2. (1) C



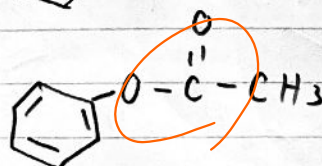
D



E



F

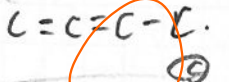
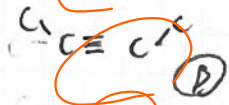
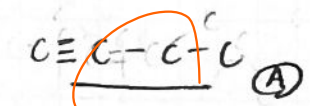


(2) 濃石炭酸

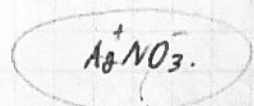
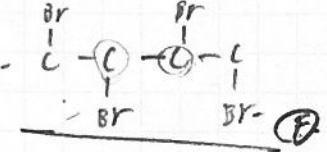
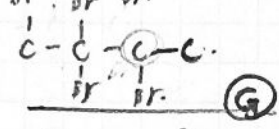
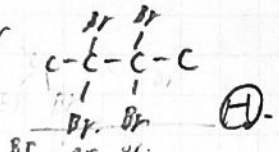
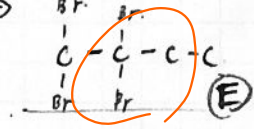
2-14

(a) A, B, C, D, E

C_4H_6

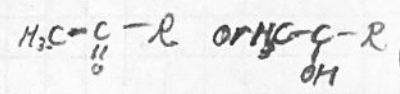


+ 2 Br₂



Aの試

水加和



4-12 C_7H_8O



問1 操作1で発生した気体 H_2

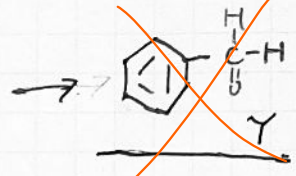
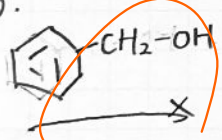
→ 金属ナトリウム反応 $-OH$ or $-COOH$

問2 フェニル類の検出

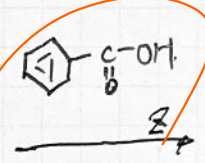
塩化鉄(III) $FeCl_3$

青色 ~ 青紫色

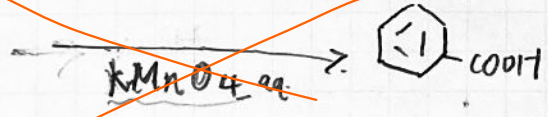
問3.



5分



問4.



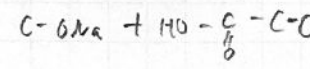
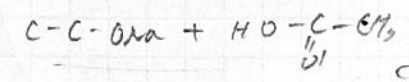
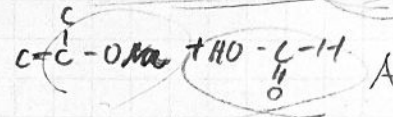
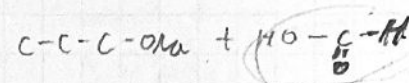
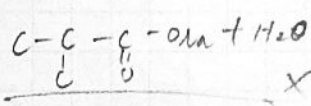
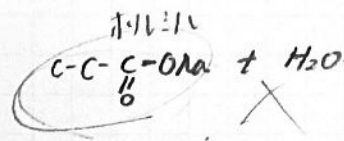
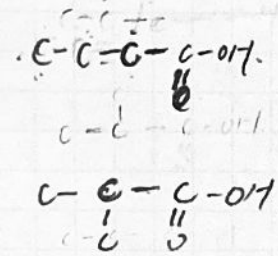
反応式

3-15
C₄H₈O₂

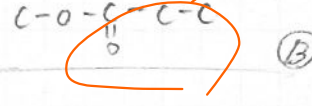
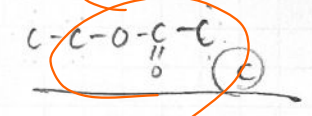
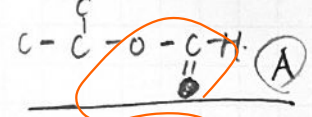
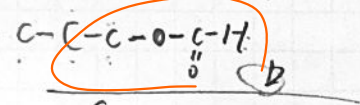
カルボン酸エステル

NaOH 中和

カルボン酸
A B C CH₃-C(=O)-CH₂-C(=O)-CH₃



72%



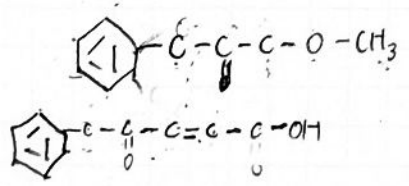
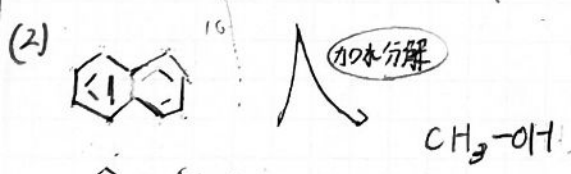
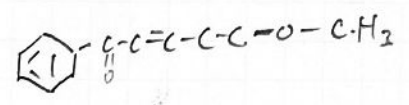
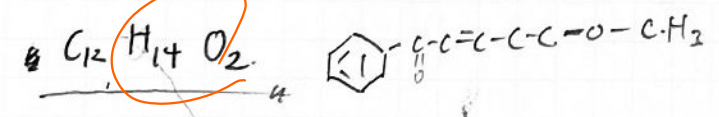
4-18

(1) C H O
75.8% 7.4% 16.8%

$\frac{75.8}{12} : \frac{7.4}{1} : \frac{16.8}{16}$

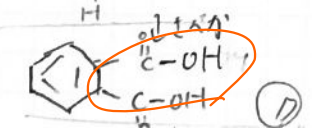
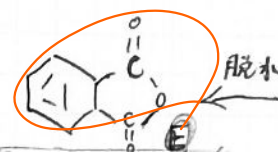
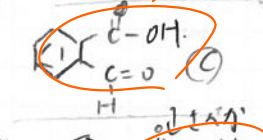
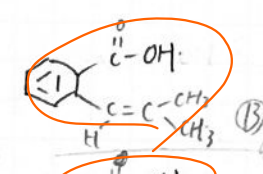
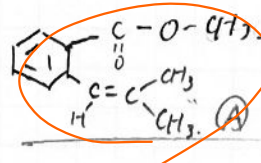
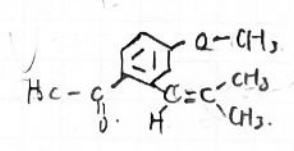
$6.15 : 7.4 : 1.05 \quad 19$
 $153 : 178 : 216 \quad 2 \times 7 + 16 = 95$

$14 : 15 : 2 \quad \frac{48}{168} \times 200 = 200$
 2×5



$U = \frac{1 \times 2 + 2 \times 14}{2} = 6$
 $\text{C}_6\text{H}_5 \sim 42$
 $\text{H}_2\text{C} \sim 2$

オゾン分解 → $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$



2-②

$$\frac{114}{160} = 0.025 \text{ mol} \quad \frac{3.21}{214} = 0.015 \text{ mol}$$

CH_2

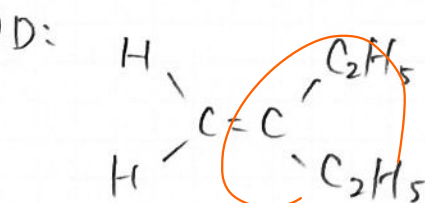
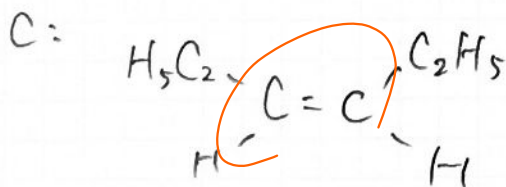
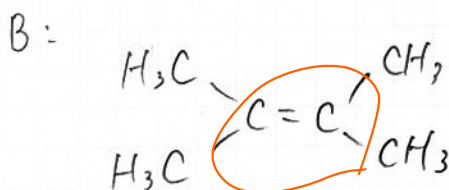
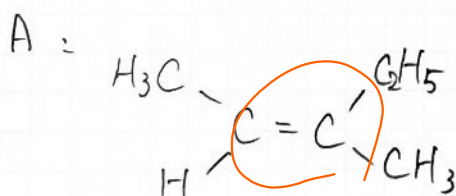
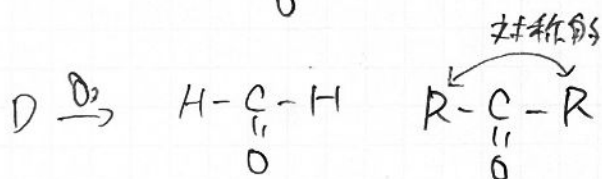
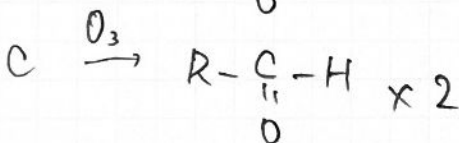
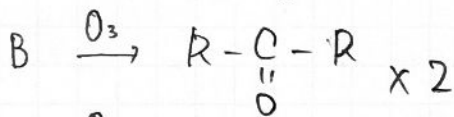
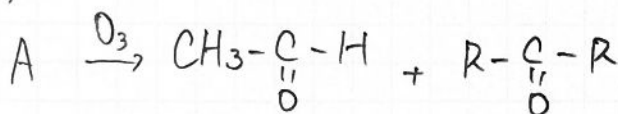
$$\frac{0.15}{n} \text{ mol of } (\text{CH}_2)_n \text{ is } 0.025 \text{ mol}$$

of Br_2 is added.

$$\Rightarrow \frac{0.15}{n} = 0.025 \quad \therefore n = 6$$

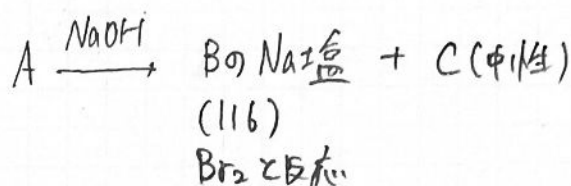


(2)



(3) A, C

3-⑦



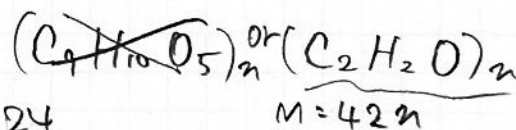
C (7112-11X) (35X) (30)

$$(1) \text{C} : \text{H} : \text{O}$$

$$= \frac{53.8}{12} : \frac{5.1}{1} : \frac{41.1}{16}$$

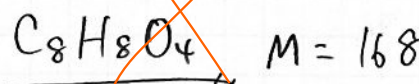
$$= 4.5 : 5.1 : 2.5$$

$$= 9 : 10.2 : 5$$



$$\frac{24}{16} = \frac{3}{2}$$

$$n = 4$$



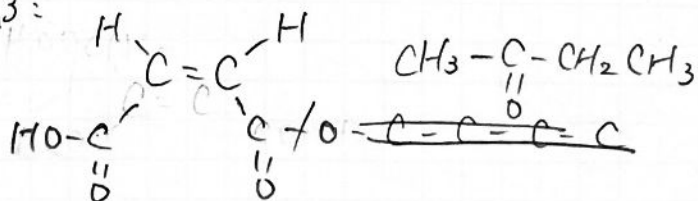
$$\begin{array}{r} 4.4 \\ 12 \overline{) 53.8} \\ \underline{48} \\ 5.8 \end{array} \quad \begin{array}{r} 2.5 \\ 16 \overline{) 41.1} \\ \underline{32} \\ 9.1 \\ \underline{80} \\ 11 \end{array}$$

$\frac{12}{9} = \frac{4}{3}$
 $\frac{168}{90} = \frac{28}{15}$
 $\frac{198}{15} = 13.2$

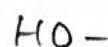
(2)

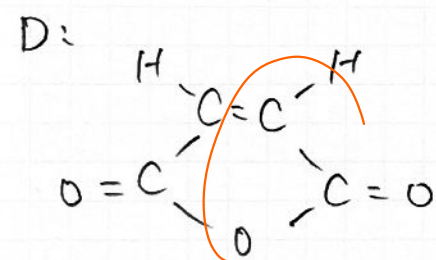
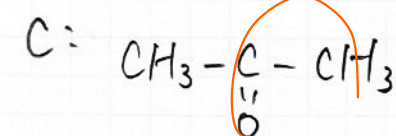
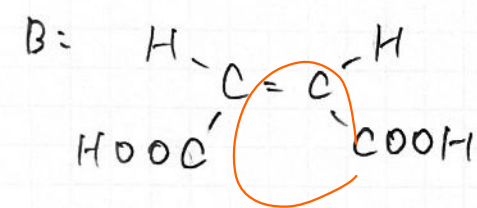
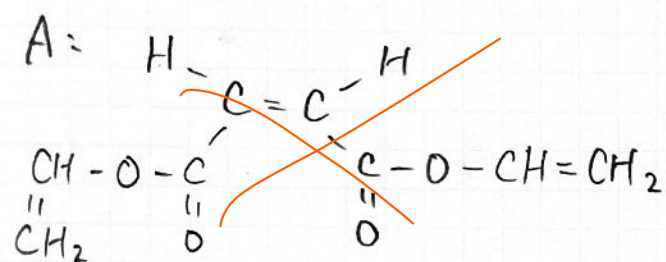
$$\textcircled{X} = \frac{16 + 2 - 8}{2} = 5$$

B:

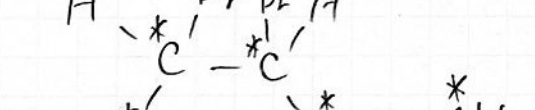


$$48 + 64 + 4 = 116$$



~~$$\begin{array}{c} \text{H} \quad \quad \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{HOOC} \quad \quad \text{C} - \text{O} - \text{C} \\ \quad \quad \quad \parallel \quad \parallel \\ \quad \quad \quad \text{O} \quad \quad \text{O} \end{array}$$~~

(3)



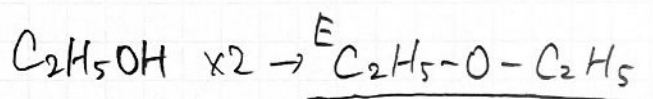
~~$2^6 = 64$~~

$$C_4H_{10}O \quad \textcircled{+} = \frac{8+2-10}{2} = 0$$

В (")

c (")

D (") $\xrightarrow[\Delta]{H_2SO_4}$ F (2重結合あり)

$$E(\text{アルコール})$$


A $\text{CH}_3(\text{CH}_2\text{CH}_2)_4\text{OH}$

B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

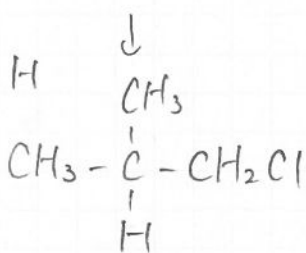
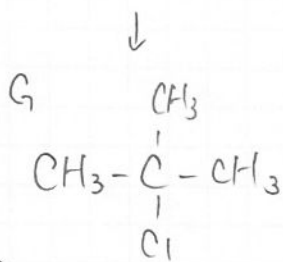
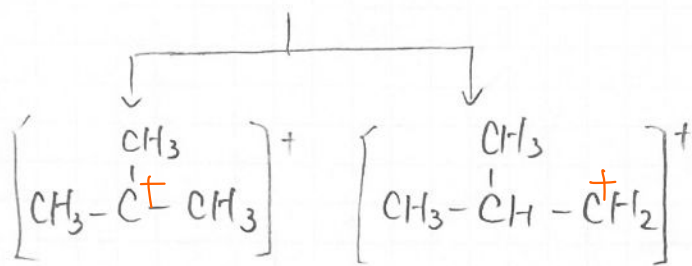
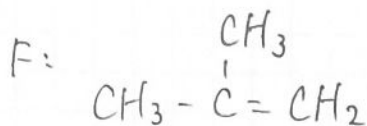
D $\text{CH}_3 - \overset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}} - \text{CH}_3$

c $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_2\text{CH}_3$

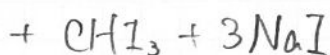
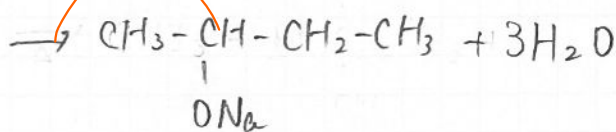
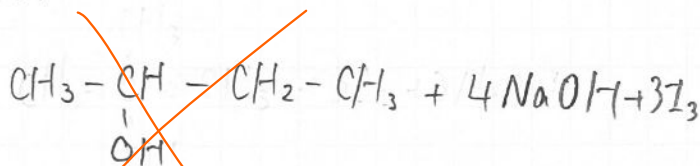
$$E \text{ } C_2H_5 - O - C_2H_5$$

3-10

(2)



(3)

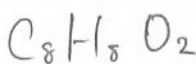


(4)

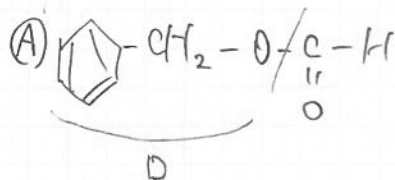
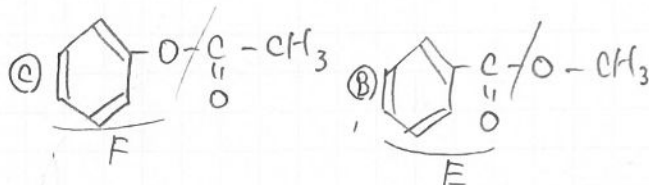
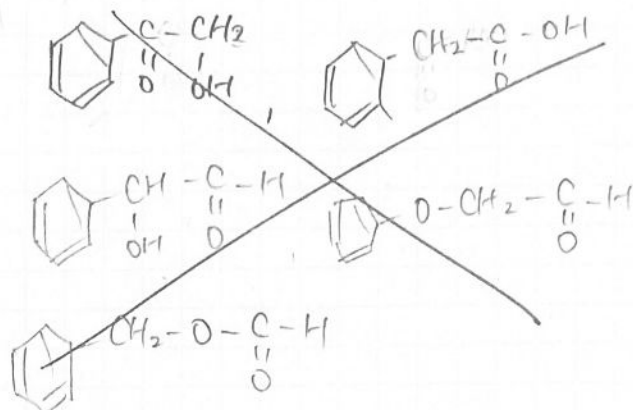
~~B~~

(5) 1-716 88/6

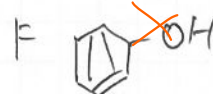
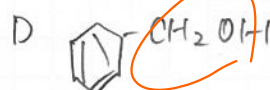
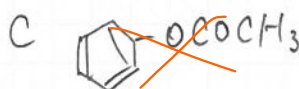
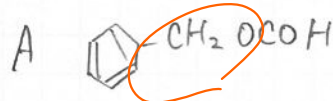
4-7



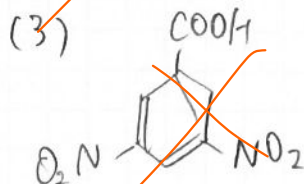
$\textcircled{7} = \frac{16+2-8}{2} = 5$



(1)

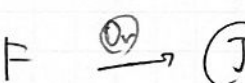
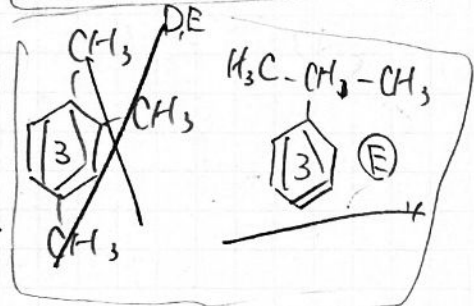
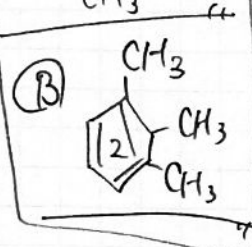
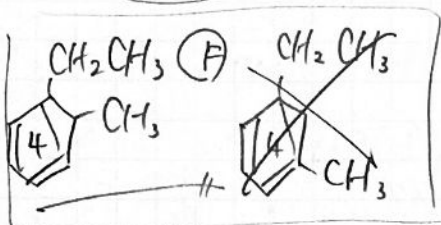
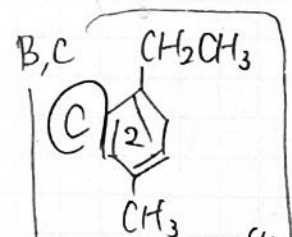
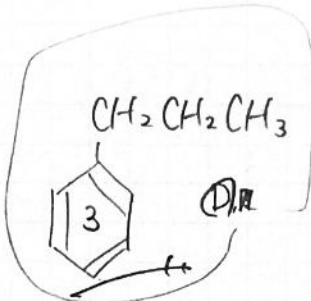
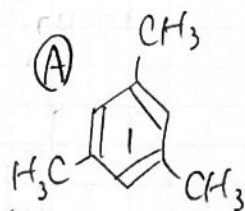


(2) ~~F~~



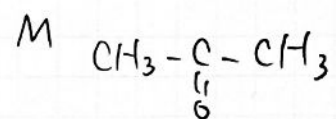
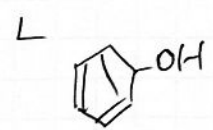
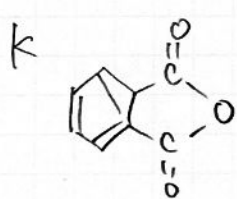
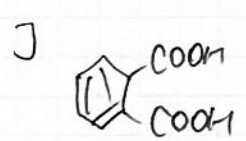
3,5-ジニトロ安息香酸

4-10



異置換体

同置換体



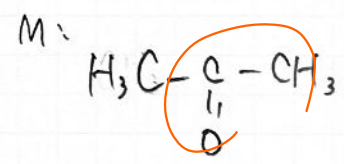
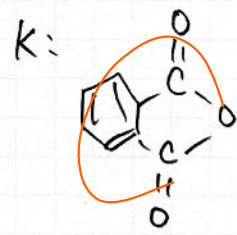
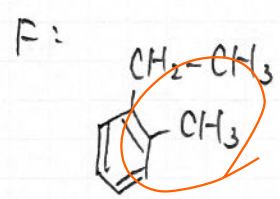
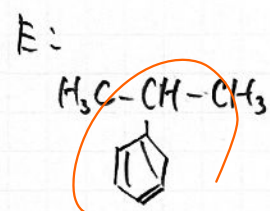
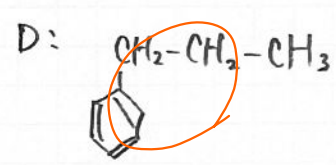
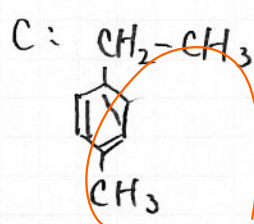
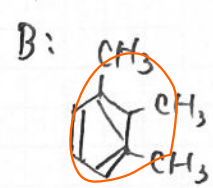
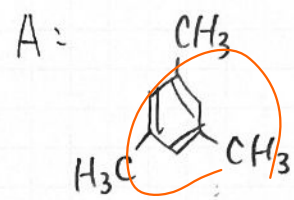
(1)

(A): 1,2-エタンジオール

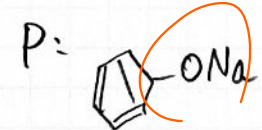
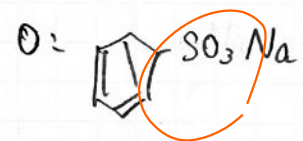
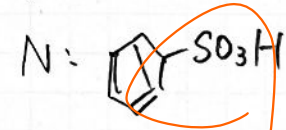
(K): 酸素

(A): 酸素

(2)



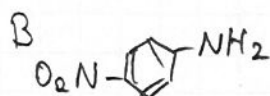
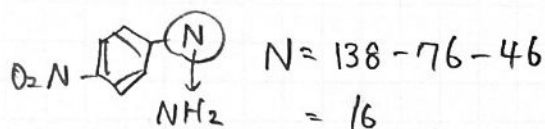
(3)



4-20

A: (カルボン酸)

$$\textcircled{A} = \frac{22 + 2 - 12 + 2}{2} = 7$$



(1) $C : H : N : O$
 $= 132 : 12 : 28 : 80$

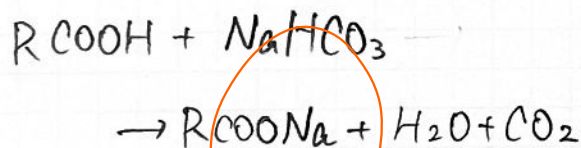
$$\begin{array}{r} 132 \\ 12 \\ 28 \\ + 80 \\ \hline 252 \end{array}$$

C ... 52.4% O ... 31.7%

H ... 4.8%

N ... 11.1%

(2)

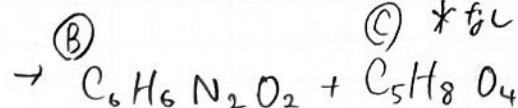
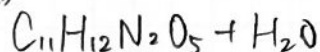


(3)

$n = 4 + 1 + 2$
 n カル

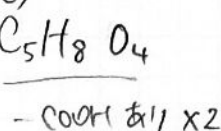


A * 17

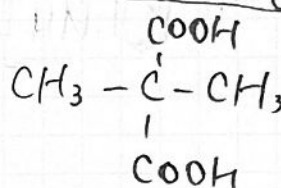
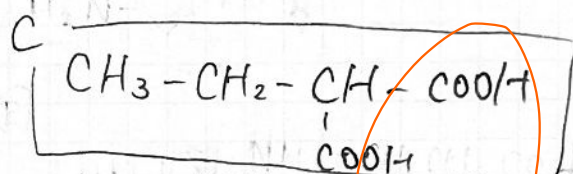
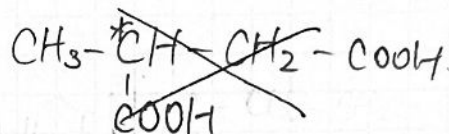
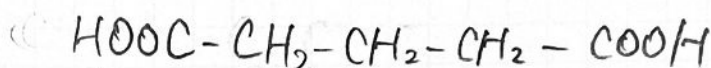
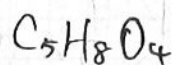


* 16

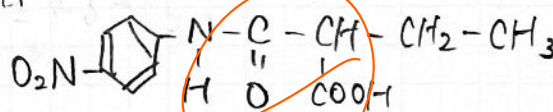
C * 16



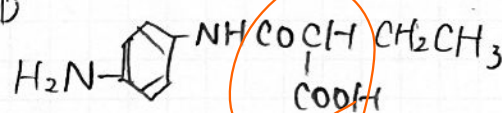
$\textcircled{C} = \frac{10 + 2 - 8}{2} = 2$



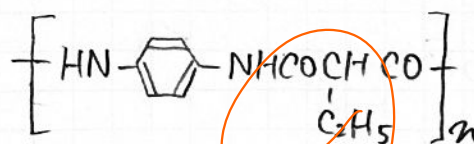
A



D



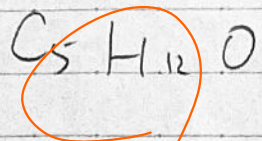
E



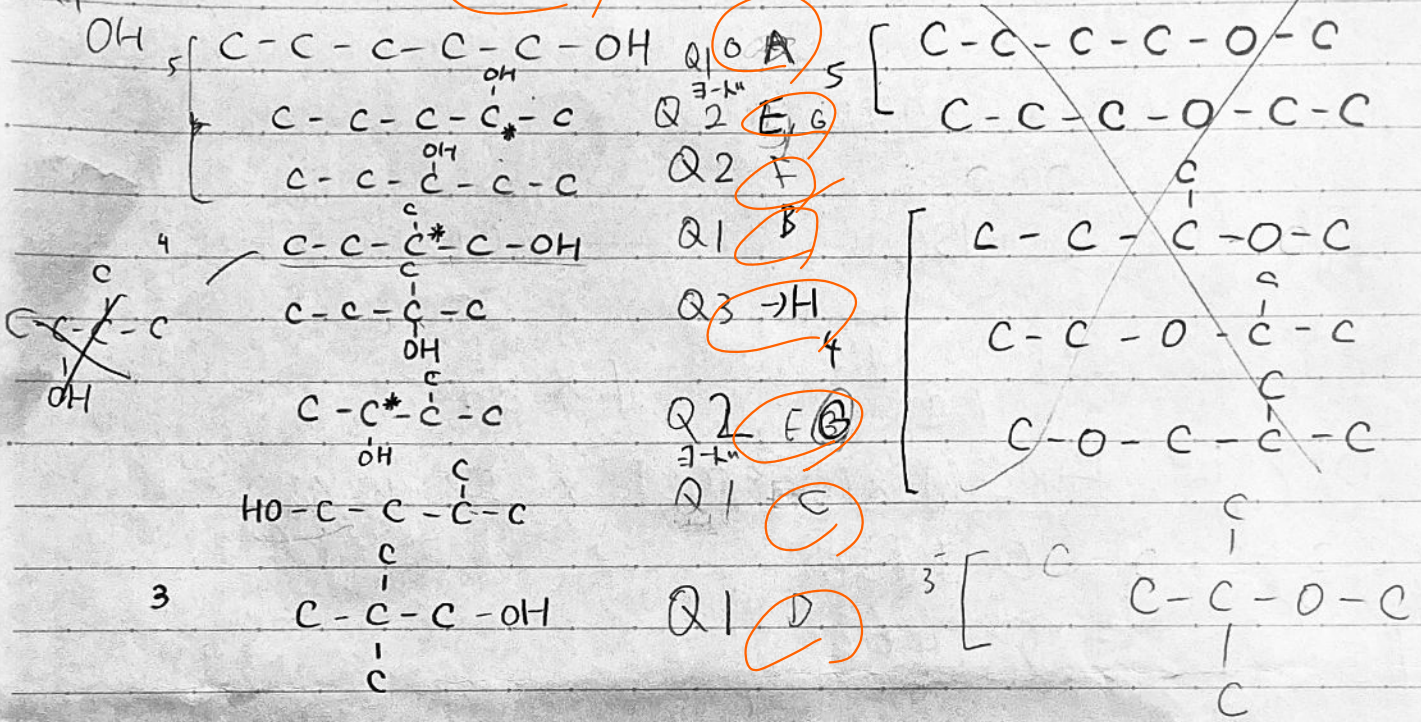
(4)

アミド結合が塩酸により加水分解
 してほろろ。

3-2

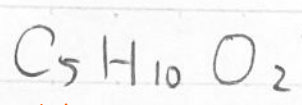


1)



(2) 6

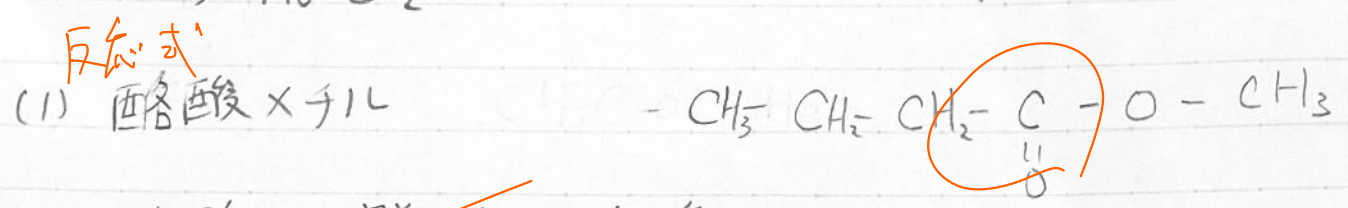
0
3-15



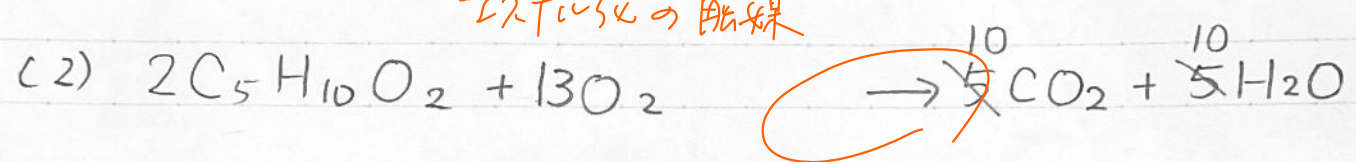
分子量
60 10 132

No. 0.962
106 | 1020
959
660
636
240

96
x 13
288
96
1248



硫酸: 脱水の促進のため。
 エステル化の触媒

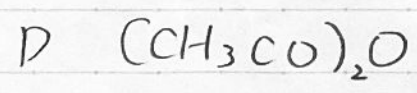
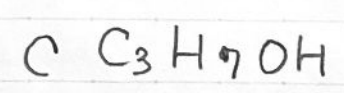
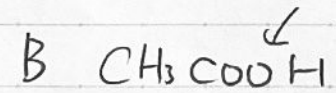
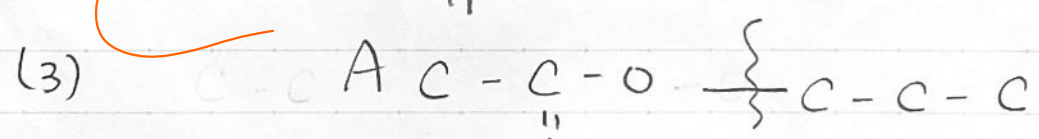


エステル x 4L $\rightarrow \frac{1.02 \times 10^{-1}}{102} = 1 \times 10^{-3} \text{ mol}$

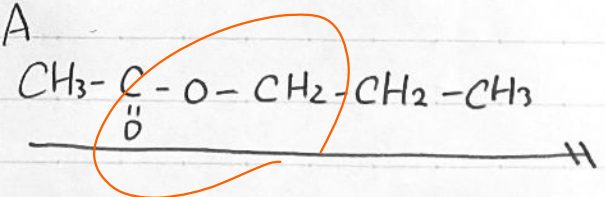
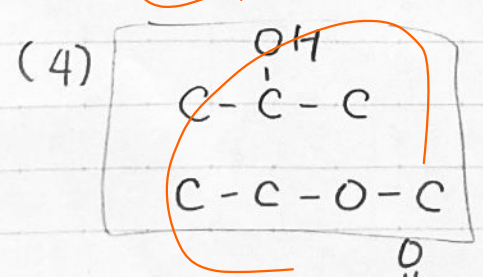
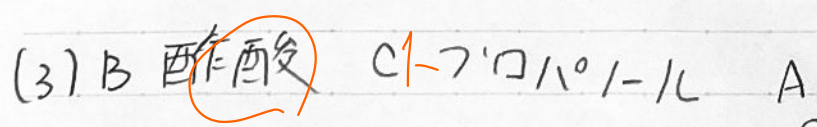
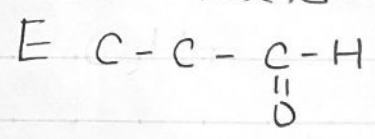
$O_2 (L) = \frac{13}{2} \times 0.1 \times 10^{-3} \text{ mol} \times 22.4 \text{ L/mol}$

$= 0.1456 \text{ L}$
 $\approx 0.146 \text{ L}$

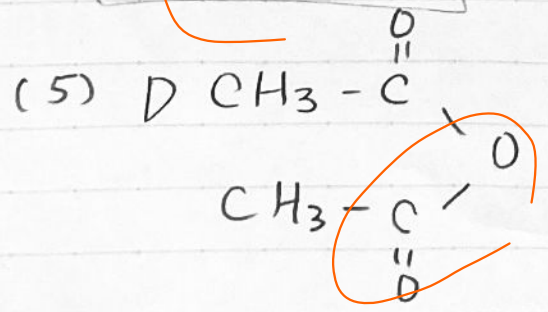
224
x 13
672
224
02912



↓ 酸化



(6) 水酸基



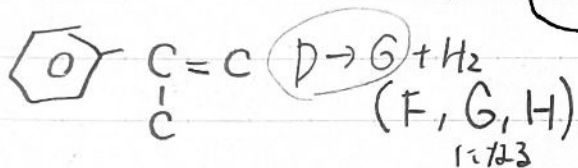
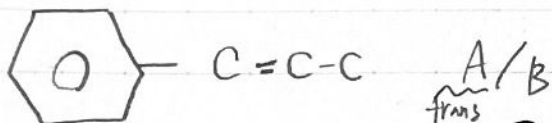
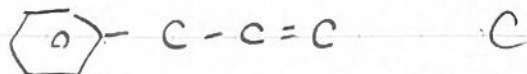
4-2

C₃H₄

C₉H₁₀

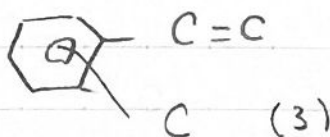
①

C₃H₅ (≡C≡)

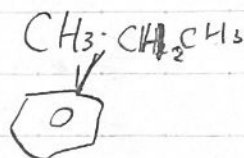


②

C₃H₆



CH₂=CH-CH₃

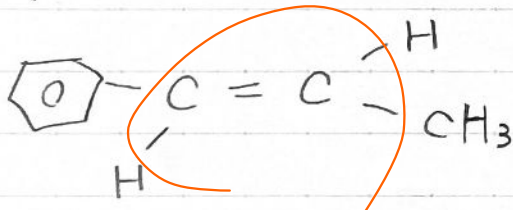


教方ありあかた
異性体あり。

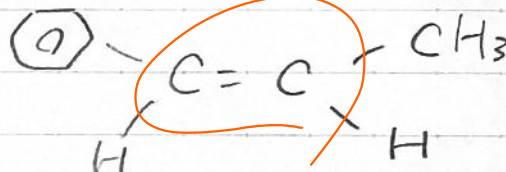
P
E
↓
H

(1) ~~6種類~~

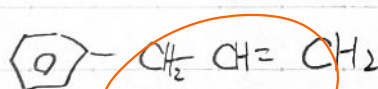
(2) A



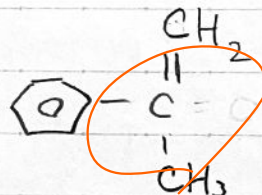
B



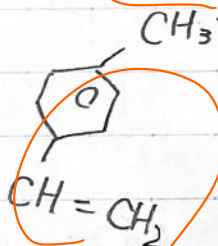
C



D



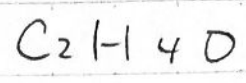
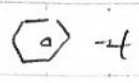
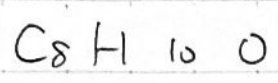
E



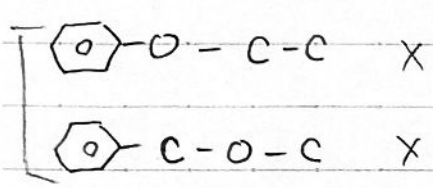
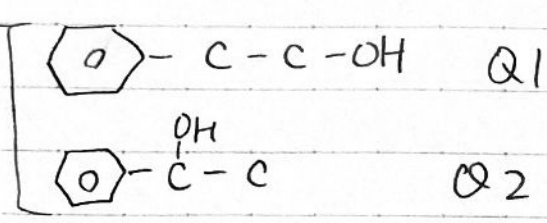
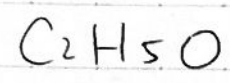
(3) 6 + 8 + 2 = 16

6 + 10 + 6 = 22

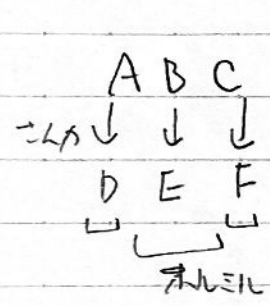
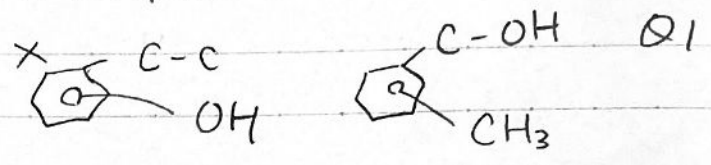
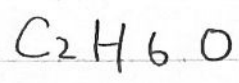
4-15



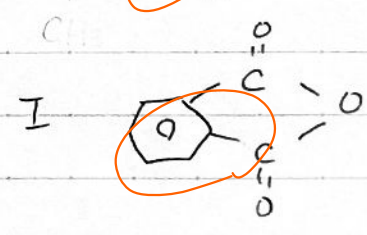
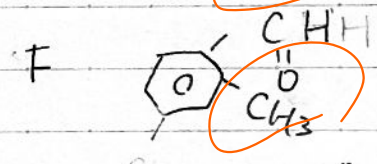
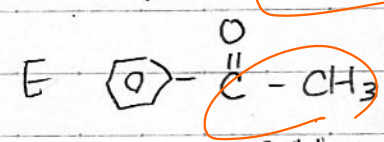
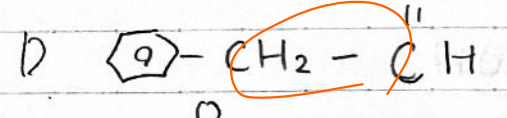
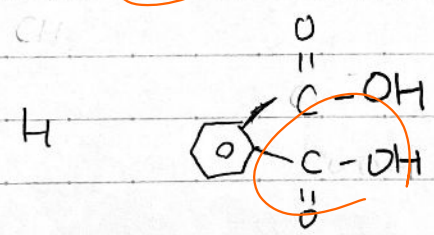
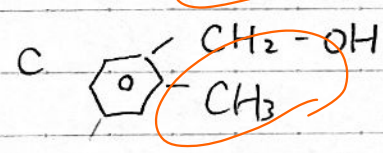
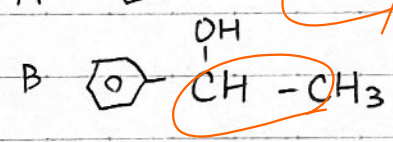
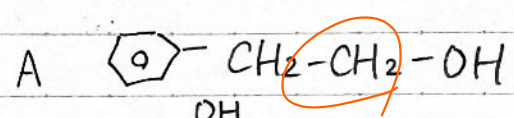
①



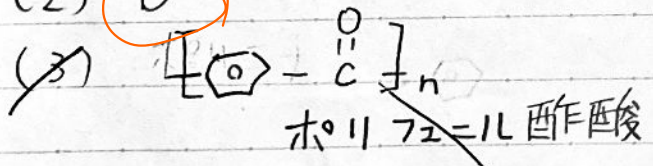
②



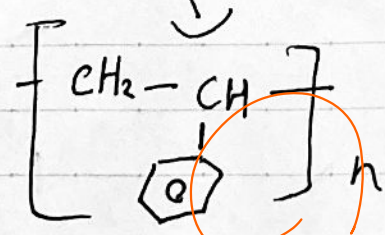
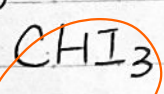
(1)



(2) B



(4) B



ポリスチレン

(脱水 → 濃硫酸 + 加熱)
 KMnO_4 + 加熱
側鎖がカルボン酸に

まさか
理由

3-112 $C_5H_{12}O$ オルアルコール

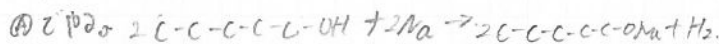
○オオ育炭素 BDF
ABC 枝が少特し。

DFヨード付

EOの脱離に付

ABの脱離に付 銀 $AgNO_3$ 付成。
ホルミル器。

問2.



④ $< Na$

Na. $\frac{0.3}{23} = 0.013$ mol

$\frac{10}{16+12+60} = 0.125$

0.013 mol

$\frac{0.3}{23} \times \frac{1}{2} \times 22.4 = 0.736$

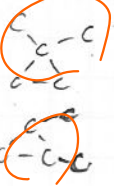
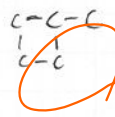
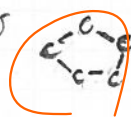
$\frac{10}{16+12+60} = 0.146$

$\div 0.15 L$

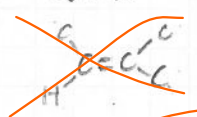
問3. 銀鏡反応



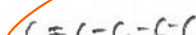
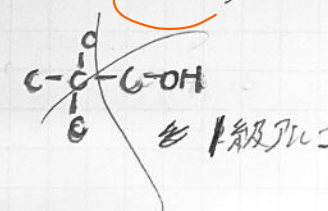
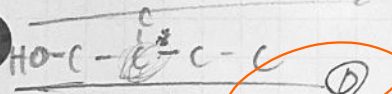
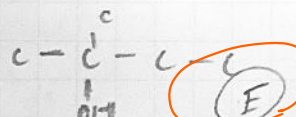
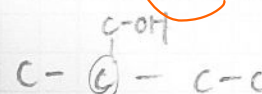
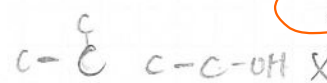
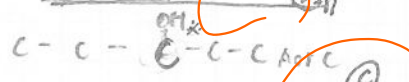
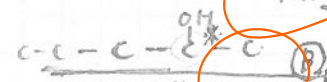
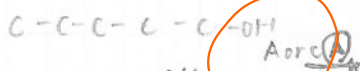
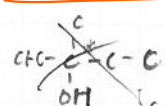
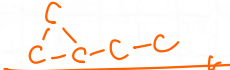
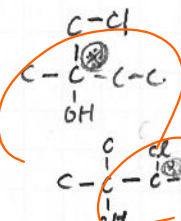
問5



問4. C_5H_{10}

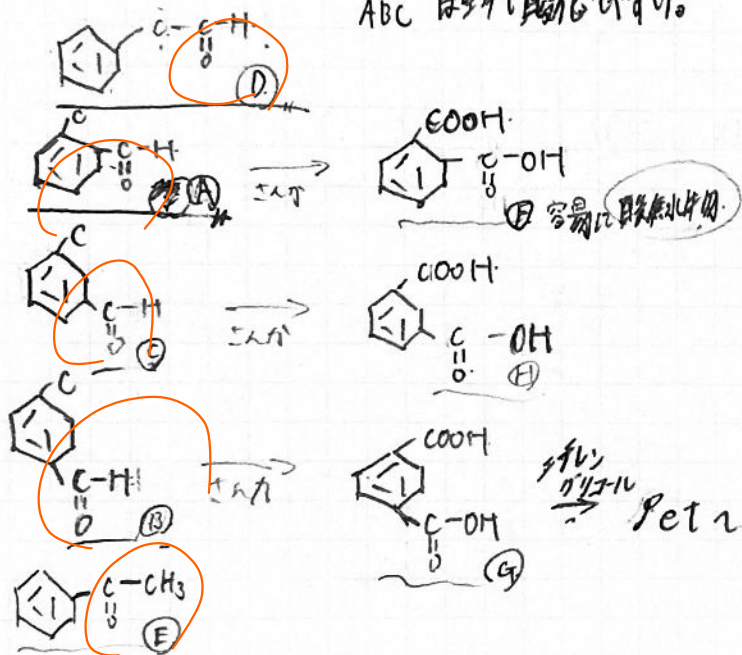


問6.



4-5. C_6H_8O カルボニル基 $-C(=O)-$

ABC はそれぞれ酸化しやすい。



F, H は官能基の配置から分子内

水素結合が起るから G は H の配置であり、
他の分子と水素結合を起しやすから。

G は F, H より

分子間水素結合が、

容易に起るから

→ 融点高い

4-1107

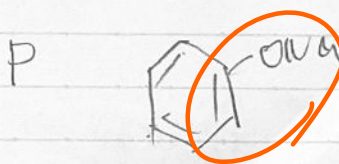
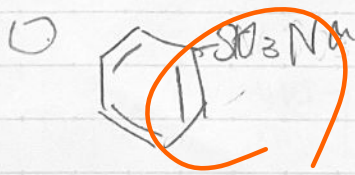
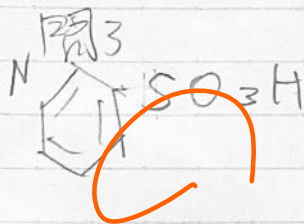
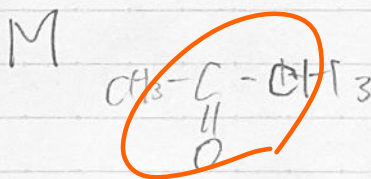
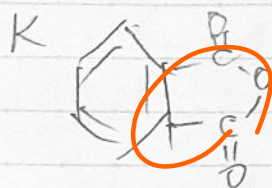
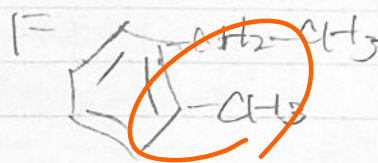
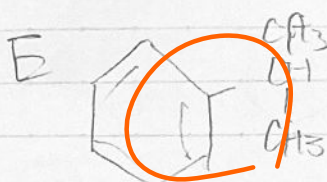
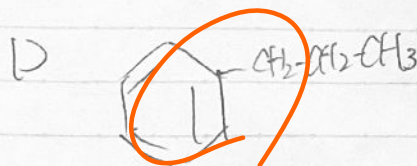
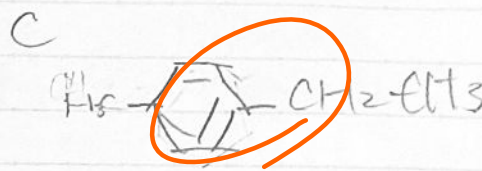
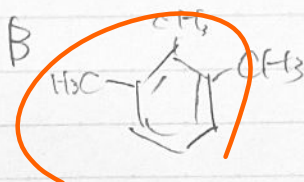
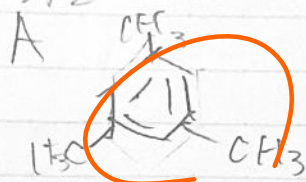
25/11

× 742-10

个酒

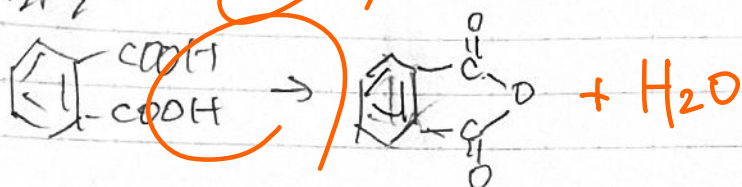
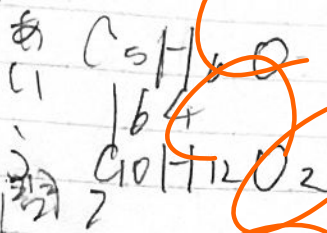
5-24

附 2

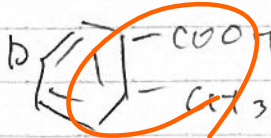
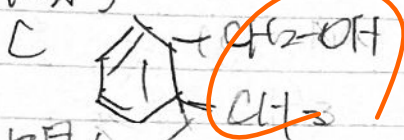


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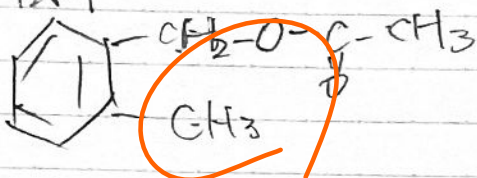
問1



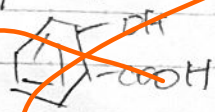
問3



問4

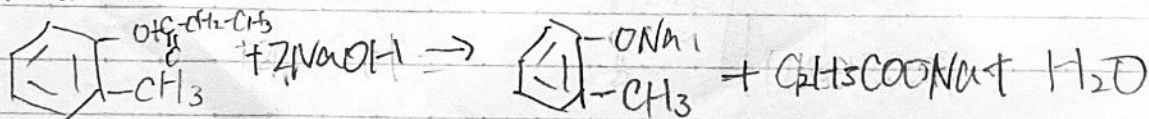


問5

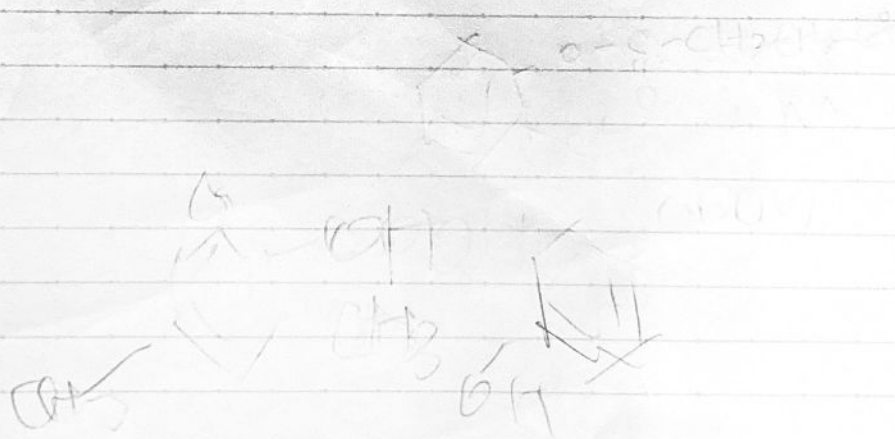


105位のやつです

問6

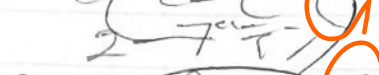
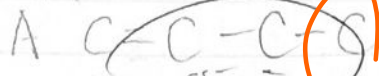


105位のやつです

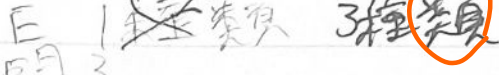


2-13

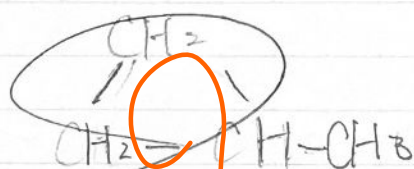
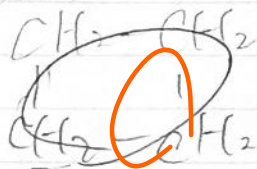
問1



問2

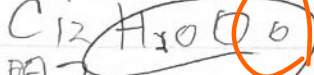


問3

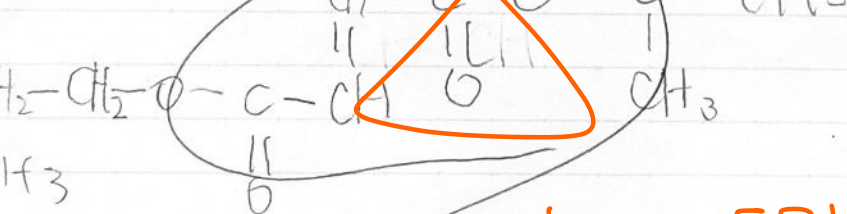
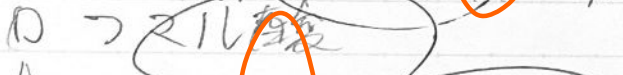
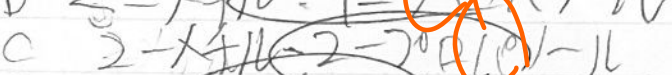


3-12

問1



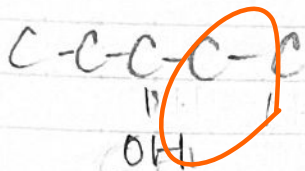
問2



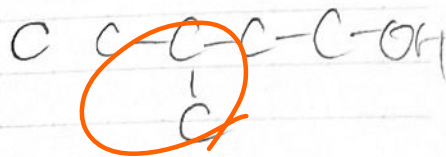
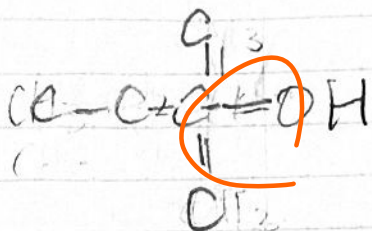
シス・トランス区別

答えはトランス型

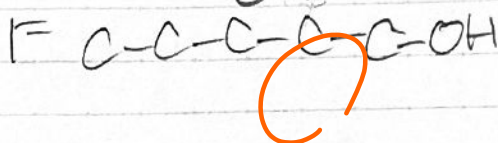
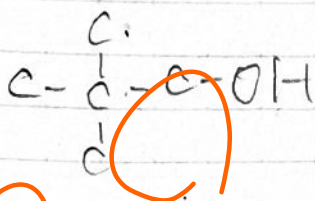
3-~~11~~
11) A



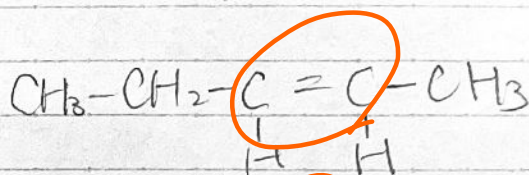
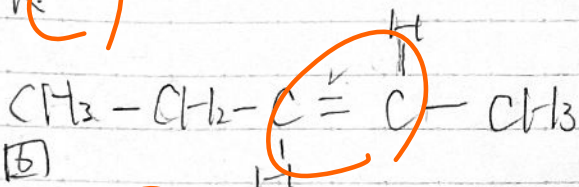
B



D

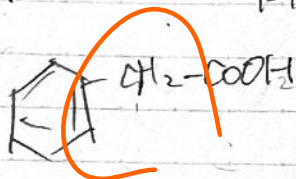


6)
(3) K

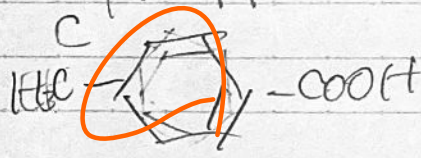
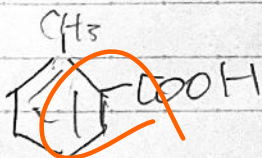


4-~~11~~

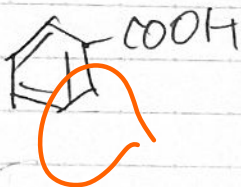
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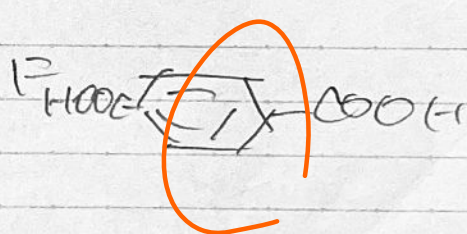
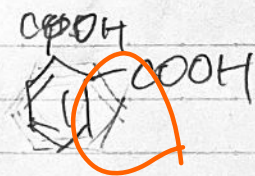
B



P



E



A

2

1